

Self-Learning: Address Decoding Techniques & 8259 Cascading

MPCA Module 4

> **Definition:** Address Decoding Techniques define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while **8259 Cascading** expands interrupt handling capacity from 8 to 64 priority channels.

1. Absolute vs Partial Address Decoding

ABSOLUTE ADDRESS DECODING:

- On a given address line, only one device is selected (Active Low).

PARTIAL ADDRESS DECODING:

- Multiple devices can be selected on the same address line using OR gates.

Comparison Table:

Feature	Absolute Decoding	Partial Decoding
Number of Devices	1	Multiple
Address Lines	Full	Partial
Complexity	Low	High
Cost	Low	High

2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one **Master 8259** is connected to up to **8 Slave 8259s** via the 3-bit Cascade Bus (**CAS_0, CAS_1, CAS_2**).

MASTER 8259: Receives interrupt on its IR line and signals Slave 8259s via Master's IR pin.

Cascading Operation Flow:

- Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.
- Master 8259 asserts its CAS_0, CAS_1, and CAS_2 lines to the Slave 8259.

3. Quick Recall Flow

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